TÉCNICO
LISBOA

Glasma Role in Jet Quenching Effects

José Soares Trindade Nunes dos Santos

Thesis to obtain the Master of Science Degree in

Physics Engineering

Supervisor(s): Prof. Full Name 1
Dr. Full Name 2

Examination Committee

Chairperson: Prof. Full Name 1
Supervisor: Prof. Full Name 2
Member of the Committee: Prof. Full Name 3

Month Year

Dedicated to someone special...

Declaration

I declare that this document is an original work of my own authorship and that it fulfills all the requirements of the Code of Conduct and Good Practices of the Universidade de Lisboa.

Acknowledgments

A few words about the university, financial support, research advisor, dissertation readers, faculty or other professors, lab mates, other friends and family...

Resumo

Inserir o resumo em Português aqui com um máximo de 250 palavras e acompanhado de 4 a 6 palavras-chave.

Palavras-chave: palavra-chave1, palavra-chave2, palavra-chave3,...

Abstract

Insert your abstract here with a maximum of 250 words, followed by 4 to 6 keywords.

Keywords: keyword1, keyword2, keyword3,...

Contents

List of Tables

List of Figures

Glossary

DIS Deep Inelastic Scattering i, 2

PDF Parton Distribution Function i, xvii, 2, 3

QCD Quantum Chromodynamics i, 3

Glossary

DIS Deep Inelastic Scattering i, 2

PDF Parton Distribution Function i, xvii, 2, 3

QCD Quantum Chromodynamics i, 3

Chapter 1

1.1 [A picture of nuclear matter]

[Part of it may migrate. At some point before, introduce the proton as something more than 3 quarks] [Add references: Thomson chapter 8 and PDG]

Probing hadrons at high energies in scattering experiments has allowed exploring the interior structure of nucleons. Data from both fixed-target and collider (HERA, Tevatron, LHC) experiments has been compiled in order to obtain pictures of the parton content of hadrons.

At HERA, electrons (or positrons) and protons were accelerated to collide at center of mass energies of around $\sqrt{s} = 300\,\mathrm{GeV}$. In some of these collisions, the momentum transfer Q^μ from the electron to the proton via a virtual photon was large enough ($Q^2 > 200\,\mathrm{GeV}^2$) to disintegrate the proton into a shower of hadrons ($e{+}p \to e{+}X$); these are called Deep Inelastic Scattering (DIS) collisions, whose cross-sections are valuable in unveiling the proton's parton content; the greater the virtual photon's energy, the greater its ability to resolve structures of smaller scale than the proton radius and with shorter characteristic timescales. DIS collisions have also been studied for this purpose using muons, μ, and neutrinos, ν, as probes, and neutrons as targets. At the Tevatron and LHC colliders, data from $p\overline{p}$ and pp collisions, respectively, has contributed to the study of the gluon content of protons, specifically.

The contributions from these many experiments have permitted the measurement of the proton's PDFs, which quantitatively characterize the parton content of a hadron. They translate the abundance, inside that hadron, of a given parton species carrying a certain fraction of its momentum (in a frame of reference where the proton's momentum is large). The PDF of a species of parton q inside the proton, $q(x)$, is defined such that the number of q partons carrying a fraction of the proton's momentum between x and $x + \Delta x$ is given by $q(x)\Delta x$. These functions actually also depend on the energy scale of the process used to probe the hadron, Q^2, as a greater exchange of energy allows probing finer features of the hadron, increasing the PDFs at small x, for example.

Fig. 1.1 shows the different parton species' PDFs for the proton. For $0.1 < x < 1$, the "valence" quarks u and d dominate – in the relative proportion expected from the static quark model for the proton, uud. However, for $x < 0.02$, they are overtaken by a "sea" of quarks and antiquarks, which, for the same flavor, have identical PDFs. For small x, however, the most abundant parton is, by far, the gluon.

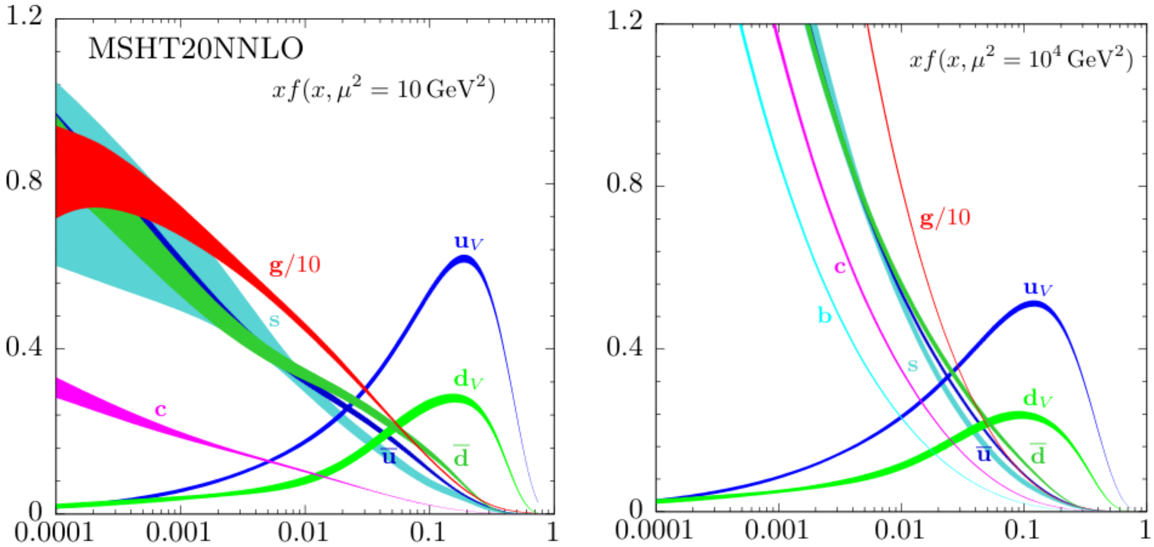

Figure 1.1: The proton's PDFs multiplied by x for the case of unpolarized beams at the energy scales $\mu^2 = 10\,\mathrm{GeV}^2$ (left) and $\mu^2 = 10^4\mathrm{GeV}^2$ (right). The gluon's PDF appears scaled down by a factor of 10 [1] Originally from and calculated by the MSHT Group. [2]

1.2 Gluon Saturation

[Refer previous discussion of low-x gluon abundance]

The splitting function of non-Abelian gauge bosons such as the gluon is a well know Quantum Chromodynamics (QCD) result. It returns the probability of emission by a parton of a gluon with relative transverse momentum $\mathbf{k}_\perp$ and a fraction x of its longitudinal momentum. At lowest order in α_S and for small x, it reads [3]:

$$\mathrm{d}P \propto \frac{\alpha_S}{\pi^2}\frac{\mathrm{d}^2\mathbf{k}_\perp}{k_\perp}\frac{\mathrm{d}x}{x} \tag{1.1}$$

the singularity of this probability as $x \to 0$ suggests a picture of abundance of small-x gluons within all hadrons, which grows uncapped as partons emit small-x gluons, which themselves emit smaller-x gluons or produce quark-antiquark pairs which also serve as gluon emitters. However, gluon densities may not explode as $x \to 0$, as unitarity would be violated. [3]

Below a certain value of x, non-linear dynamics not considered so far should take over and tame the divergence, *saturating* the low-x gluon density in the hadron. The onset of saturation is controlled by the saturation scale, $Q_s(x)$, a transverse momentum scale below which non-linear effects become relevant. [3]

One interpretation of these non-linear effects is the onset, above certain gluon densities, of gluon recombination. Above that threshold, the probability of emission of a new gluon becomes comparable to the probability of recombination of gluons, stopping the growth of their density (Fig. 1.2).

More formally, the BFKL (Balitsky-Fadin-Kuraev-Lipatov) QCD evolution equation is a linear differential equation that governs how gluon density in momentum space within a hadron should evolve with x when that hadron is probed at a certain energy scale. It is a linear equation, and it predicts a di-

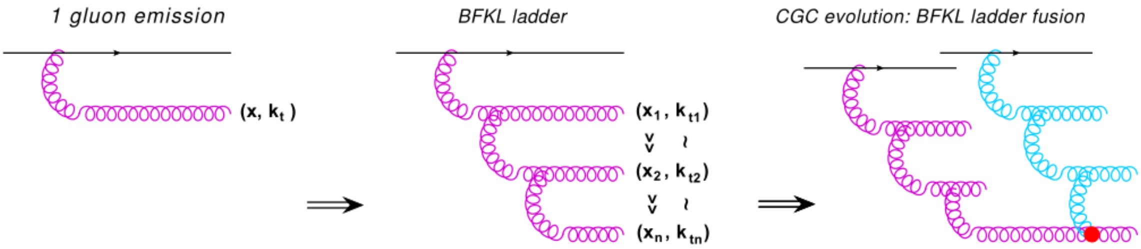

Figure 1.2: The onset of gluon recombination. Above a certain gluon density, gluon recombination (red dot) becomes comparatively likely to gluon emission (the pink and blue ladders of gluons). Taken from [3].

vergence of the gluon density as $x \to 0$. The equation does not contain terms reflecting interference between emitters of gluons, as it assumes the hadron to be a dilute parton system. However, making the probability of emission of a new gluon dependent on the present gluon density allows for saturation to be reached when gluon generation and gluon recombination balance out. QCD evolution equations which attempt to incorporate this effect have additional, non-linear, density-dependent terms, such as the GRL-MQ equation (Gribov-Riskin-Levin-Mueller-Qiu). [3]

1.3 The Color Glass Condensate

The Color Glass Condensate (CGC) effective theory of QCD for high-energy QCD scattering is one of the approaches developed to study initial state effects in HICs, and has stood out due to its practical applicability and phenomenological success. [3]

[INCOMPLETE]

1.4 The MV model

1.5 [Setting the stage/introducing the initial situation]

In a HIC, as previously introduced, two heavy nuclei travel in opposite directions at near the speed of light before crashing into each other. For the purposes of this work, both nuclei will be taken to travel at the speed of light; because of this, they suffer length contraction along the direction of their trajectories, becoming infinitely thin/planar. Furthermore, their collision is taken to be central, that is, the trajectories of their centers are perfectly aligned.

In the laboratory/center of mass frame, the x^1 axis is placed along the trajectories of the nuclei. The nucleus moving in the positive x^1 direction is identified as nucleus 1, and the one moving in the negative x^1 direction is identified as nucleus 2. The nuclei collide at $x^1 = 0$, at instant $t = 0$. This system is invariant under boosts along the x^1 direction.

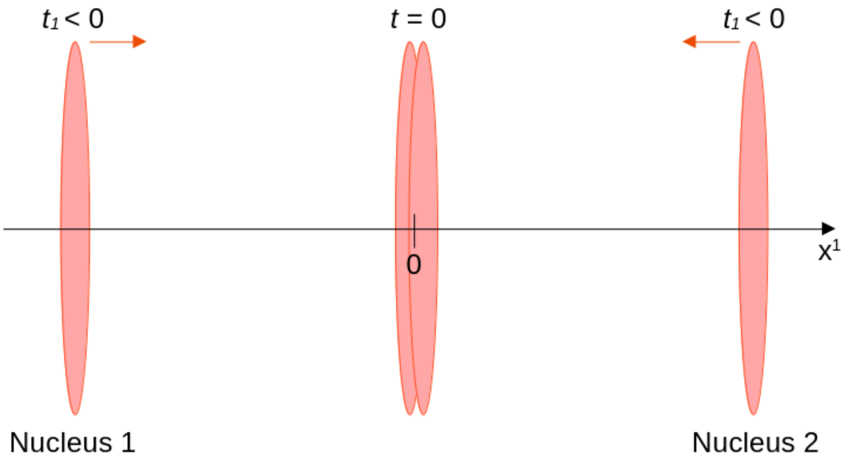

Figure 1.3: Two heavy nuclei, Lorentz-contracted into thin disks, travel along the x^1 axis in opposite directions until they collide at $t = 0, x^1 = 0$.

As such, prior to the collision, each nucleus's position four-vector, $x^\mu = (x^0, x^1, x^2, x^3)$, in Minkowski coordinates, is:

$$\text{Nucleus 1}: \quad (t, t, 0, 0)$$
$$\text{Nucleus 2}: \quad (t, -t, 0, 0)$$

These trajectories are well-illustrated in the x^1Ox^0 plane:

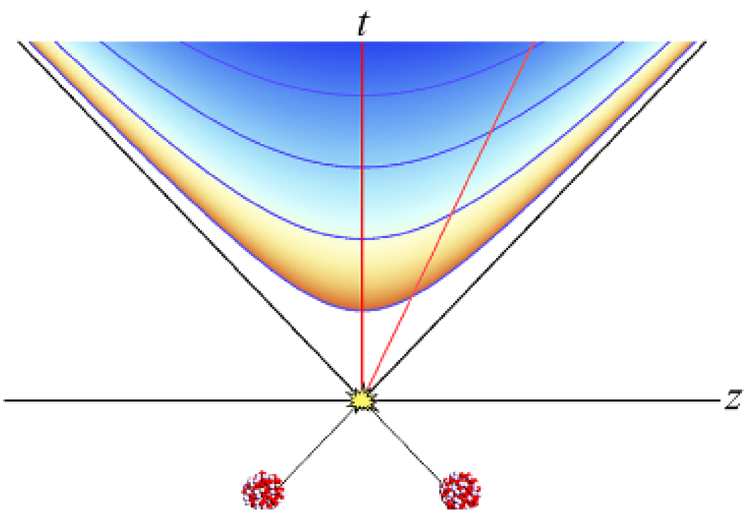

Figure 1.4: A heavy-ion collision represented in the x^1Ox^0 plane. Both nuclei travel along the light-cone centered around the collision at coordinates $0, 0)$. Taken from [4] The reference for this.

After the collision, the nuclei's approximately static high-x partons, mentioned in the previous section, carry on in the direction they were moving at the speed of light, acting as color sources for the gluon fields then generated in the space between the now receding nuclei [They were already acting as color sources for fields, but in the other quadrants]. It is in this space that the pre-hydrodynamic stage to be studied is formed. Fig. 1.4 already suggests the usefulness of the light cone coordinate system (see Appendix

??), as the static color sources only travel along the edges of the light cone centered on the collision coordinates $(0,0)$. In this coordinate system, the nuclei's position four-vectors, $x^\mu = (x^+, x^-, x^2, x^3)$, are:

$$\begin{aligned} \text{Nucleus 1}: &\quad (\sqrt{2}t, 0, 0, 0) \\ \text{Nucleus 2}: &\quad (0, \sqrt{2}t, 0, 0) \end{aligned}$$

and the collision coordinates remain at $(x^+ = 0, x^- = 0)$.

In order to fulfill this work's objectives, it will be necessary to calculate the gluon fields generated in the space between the receding nuclei, making use of the classical Yang-Mills equations, discussed further ahead. The fields will be determined from the the static distributions of color charge of the receding nuclei.

Next: event averages? Move the introduction to dEdx and $\hat{q}$ to before this? say they are the result of the color fields acting on the quark probe?

1.6 The QCD (Yang-Mills?) Lagrangian

In studying a system where only strong interactions are relevant, one turns to the QCD Lagrangian. It is a functional of the quark fields, ψ, which are 4-component Dirac spinors, and of the gluon fields, A_μ^a, which are real-valued four-vectors; both are functions of spacetime, x. For a theory with N_c colors, considering a single flavor of quarks with mass m, the Lagrangian is

$$\mathcal{L}_{\text{QCD}} = \overline{\psi}\left(i\gamma^\mu D_\mu - m\right)\psi - \frac{1}{2}\text{Tr}_c\left(F_{\mu\nu}F^{\mu\nu}\right) \tag{1.2}$$

where D_μ, the covariant derivative, is given by:

$$D_\mu = \partial_\mu - igA_\mu, \quad \text{where } A_\mu = A_\mu^a T_a \tag{1.3}$$

and $F^{\mu\nu}$, the gluon field strength tensor, is given by:

$$\begin{aligned} F_{\mu\nu} &= \partial_\mu A_\nu - \partial_\nu A_\mu - ig[A_\mu, A_\nu] \\ F_{\mu\nu}^a &= \partial_\mu A_\nu^a - \partial_\nu A_\mu^a + gf^{abc}A_\mu^b A_\nu^c \end{aligned} \tag{1.4}$$

g is the Strong Force's coupling constant; T^a are the SU(N_c) generators in the fundamental representation, and f^{abc} are the SU(N_c) structure constants; Tr_c denotes a color-trace, that is, a trace in color space, where the T_a live; and $\overline{\psi}$ is the quark field's Dirac adjoint, equal to $\psi^\dagger\gamma^0$.

Writing the Lagrangian with all indices explicit,

$$\mathcal{L}_{\text{QCD}} = \left(\psi_\alpha^i\right)^*\left(\gamma^0\right)_{\alpha\beta}\left(i\left(\gamma^\mu\right)_{\beta\gamma}\left(\delta^{ij}\partial_\mu - igA_\mu^a\left(T^a\right)^{ij}\right) - \delta_{\beta\gamma}\delta^{ij}m\right)\psi_\gamma^j - \frac{1}{4}F_{\mu\nu}^a F_a^{\mu\nu} \tag{1.5}$$

where $\alpha, \beta, \gamma = 1, 2, 3, 4$ are Dirac indices, $i, j = 1, ..., N_c$ are color indices in the fundamental representation, $a = 1, ..., {N_c}^2 - 1$ are color indices in the adjoint representation, and $\mu, \nu = 0, 1, 2, 3$ are spacetime indices when using Minkowski coordinates.

To obtain the Lagrangian considering all flavors of quarks, one need only add copies of the first term of Eq. (1.2) for each quark flavor's quark field.

1.7 The Yang-Mills Equations

Given the presented QCD Lagrangian, the corresponding Euler-Lagrange equations relating to the gluon fields A are called the Yang-Mills equations, which can be thought of as the QCD equivalent to Electrodynamics' Maxwell's equations. In the absence of color sources,[1] they read:

$$[D_\mu, F^{\mu\nu}] = 0 \tag{1.6}$$

In light cone coordinates, for nucleus 1, the only non-vanishing component of the color current four-vector J^ν is the $+$ component; for nucleus 2, the only non-vanishing component is the $-$ component[2]:

$$ {J_1}^+ = g\delta(x^-)\rho_1(\mathbf{x}_\perp) \qquad {J_2}^- = g\delta(x^+)\rho_2(\mathbf{x}_\perp)$$

where $\rho_{1,2}$ are the static number color densities in nuclei 1 and 2 per unit of transverse area. The Lorentz contraction in the x^1 direction suffered by nucleus 1 implies a very thin spread of color charge in the x^- direction, so the dependence on this coordinate is factorized as $\delta(x^-)$ [6] This reference; if we consider additionally that, due to length contraction along the x^1 direction, the nucleus is infinitely thin, and therefore extends only in the $\mathbf{x}_\perp$ plane, then ρ_1 may not depend on x^+ either. Equivalently, for nucleus 2, which travels along the x^- axis, the color charge spread along the x^+ direction is factorized into $\delta(x^+)$ and, if considered infinitely thin, neither on x^-.

Adding the contributions of both color currents, the source term becomes[3]:

$$J^\nu = g\delta^\nu_+\delta(x^-)\rho_1(x^+, \mathbf{x}_\perp) + g\delta^\nu_-\delta(x^+)\rho_2(x^-, \mathbf{x}_\perp) \tag{1.7}$$

The classical Yang-Mills equations to be solved in the presence of the static color sources become, then,

$$[D_\mu, F^{\mu\nu}] = J^\nu \tag{1.8}$$

[1] That is, in the absence of source terms in the Lagrangian and taking the usual color current, $g\overline{\psi}\gamma^v\psi$, to be 0.
[2] [5] allows the nuclei to retain some thickness and then brings it to 0 at the end of the calculation. It justifies the use of ρ as a surface charge density, I think.
[3] Justify why the color-current becomes the source term?

1.8 Transcribed expressions yet without text

$$\begin{aligned} E_0^\eta(\mathbf{x}_\perp) &= -ig\delta^{ij}[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)] \\ &= g\delta^{ij}\alpha_{1a}^i(\mathbf{x}_\perp)\alpha_{2b}^j(\mathbf{x}_\perp)f^{abc}T_c \end{aligned} \tag{1.9a}$$

$$\begin{aligned} B_0^\eta(\mathbf{x}_\perp) &= -ig\varepsilon^{ij}[\alpha_1^i(\mathbf{x}_\perp), \alpha_2^j(\mathbf{x}_\perp)] \\ &= g\varepsilon^{ij}\alpha_{1a}^i(\mathbf{x}_\perp)\alpha_{2b}^j(\mathbf{x}_\perp)f^{abc}T_c \end{aligned} \tag{1.9b}$$

$$E^\eta(\tau, \mathbf{k}_\perp) = E_0^\eta(\mathbf{k}_\perp)J_0(k\tau) \tag{1.10a}$$

$$E^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}\hat{k}^j B_0^\eta(\mathbf{k}_\perp)J_1(k\tau) \tag{1.10b}$$

$$B^\eta(\tau, \mathbf{k}_\perp) = B_0^\eta(\mathbf{k}_\perp)J_0(k\tau) \tag{1.10c}$$

$$B^i(\tau, \mathbf{k}_\perp) = -i\varepsilon^{ij}\hat{k}^j E_0^\eta(\mathbf{k}_\perp)J_1(k\tau) \tag{1.10d}$$

In order to get expressions for one of the fields in coordinate space[4], the appropriate field at $\tau = 0^+$ (eqs. 1.9) must be Fourier-transformed and used in Eqs. (1.10). The inverse Fourier transform is then applied to the resulting expression. For E^α, $\alpha = 1, 2$ (2,3)?, for example:

$$\begin{aligned} E^\alpha(\tau, \mathbf{x}_\perp) = \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} e^{i\mathbf{k}_\perp\cdot\mathbf{x}_\perp} \, (-i)\varepsilon^{\alpha\beta}\hat{k}^\beta J_1(k\tau) \int \mathrm{d}^2\mathbf{u}_\perp e^{-i\mathbf{k}_\perp\cdot\mathbf{u}_\perp} \\ \times \; g\varepsilon^{ij}\alpha_{1a}^i(\mathbf{u}_\perp)\alpha_{2b}^j(\mathbf{u}_\perp)f^{abc}T_c \\ = -ig\varepsilon^{ij}f^{abc}T_c\varepsilon^{\alpha\beta} \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} \hat{k}^\beta J_1(k\tau) \int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp} \\ \times \; \alpha_{1a}^i(\mathbf{u}_\perp)\alpha_{2b}^j(\mathbf{u}_\perp) \end{aligned} \tag{1.11}$$

And the remaining fields:

$$E^\eta(\tau, \mathbf{x}_\perp) = g\delta^{ij}f^{abc}T_c \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} J_0(k\tau) \int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha_{1a}^i(\mathbf{u}_\perp)\alpha_{2b}^j(\mathbf{u}_\perp) \tag{1.12}$$

$$B^\eta(\tau, \mathbf{x}_\perp) = g\varepsilon^{ij}f^{abc}T_c \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} J_0(k\tau) \int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha_{1a}^i(\mathbf{u}_\perp)\alpha_{2b}^j(\mathbf{u}_\perp) \tag{1.13}$$

$$B^\alpha(\tau, \mathbf{x}_\perp) = -ig\delta^{ij}f^{abc}T_c\varepsilon^{\alpha\beta} \int \frac{\mathrm{d}^2\mathbf{k}_\perp}{(2\pi^2)} \hat{k}^\beta J_1(k\tau) \int \mathrm{d}^2\mathbf{u}_\perp e^{i\mathbf{k}_\perp\cdot(\mathbf{x}-\mathbf{u})_\perp}\alpha_{1a}^i(\mathbf{u}_\perp)\alpha_{2b}^j(\mathbf{u}_\perp) \tag{1.14}$$

Maybe using the abbreviation ${\alpha_{1a}^i}^{\mathbf{u}}$ or $\alpha_{1a}^{i\mathbf{u}} \equiv \alpha_{1a}^i(\mathbf{u}_\perp)$ it will be more readable?

[4] as is necessary in order to calculate $X^{\alpha\beta}$

Chapter 2

[Margaret]

[Likely to Migrate, requires references (textbooks?)]

The objectives of this work are: the derivation, by analytical means, of expressions for parameters characterizing the pre-equilibrium stage of a HIC as a medium which affects the highly energetic partons that traverse it; and the numerical evaluation of those expressions and comparison of the results obtained to others in the literature.

Those parameters are: the *stopping power*, denoted by $-\,\mathrm{d}dE/\,\mathrm{d}x$, and the *momentum broadening coefficient*, denoted by $\hat{q}$.

In Particle Physics, the stopping power of a medium as felt by a probe traversing it at a given instant corresponds to the amount of kinetic energy being lost by the probe per unit length traveled in the medium. Less formally, it is equal to the infinitesimal loss of energy of the probe by interaction with the medium divided by the infinitesimal distance traversed during that loss:

$$-\frac{\mathrm{d}E}{\mathrm{d}x} \approx -\frac{\Delta E}{\Delta x}$$

The stopping power may be viewed as a measure of how capable a medium is of slowing down or stopping traversing particles within a given length, which makes it a quantity commonly discussed in the field of radiation protection, for example.

The momentum broadening coefficient, on the other hand, measures how capable a medium is of deflecting traversing particles, that is, of changing their original direction. It corresponds to the square magnitude of transverse momentum gained by the probe per unit length traveled in the medium, transverse meaning perpendicular to the probe's velocity in that instant.

$$\hat{q} \approx \frac{(\Delta \mathbf{p}_T)^2}{\Delta x}$$

These quantities are dependent on the properties of the medium and of the probe, on the velocity of the probe, and, in the case of a heterogeneous, dynamic medium, on the current position of the probe and on time.

The chromodynamic field correlators derived in the previous section may be used to obtain the momentum broadening coefficient, $\hat{q}$, and the stopping power, $\mathrm{d}E/\,\mathrm{d}x$, felt by a relativistic quark probe traversing the glasma. This procedure is outlined in [7].

2.1 The Fokker-Planck equation and its collision terms

The Fokker-Planck equation has frequently been used to study the transport of heavy quarks across a thermalized QGP. The following Fokker-Planck equation, derived in [8] for the transport of heavy quarks through a glasma, is also applicable to relativistic light partons, as long as the diffusion approximation is applicable [7], that is, as long as the transfer of momentum to the probe is negligible when compared to the probe's momentum [8]. It commands the evolution of the event-averaged probe quark distribution function $n(t, \mathbf{x}, \mathbf{p})$ as the probes traverse the medium.

$$\left(\frac{\partial}{\partial t} + \mathbf{v}\cdot\nabla - \nabla_p^\alpha X^{\alpha\beta}\nabla_p^\beta - \nabla_p^\alpha Y^\alpha\right) n(t, \mathbf{x}, \mathbf{p}) = 0 \tag{2.1}$$

where t is time, v is the probe's velocity, $\nabla^\alpha = (\partial_1, \partial_2, \partial_3)$ is the usual gradient, ∇_p^α is the gradient with respect to the probes' momentum, $\partial/(\partial p^\alpha)$, $\mathbf{x}$ and $\mathbf{p}$ are the probes' position and momentum, respectively, and α, β run from 1 through 3. The tensors $X^{\alpha\beta}$ and Y^α, from which $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ will be calculated, are called the collision terms.

The collision terms are related to event-averaged[1] rates of change of the momentum of the probes over time:

$$\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = X^{\alpha\beta} + X^{\beta\alpha} \tag{2.2a}$$

$$\frac{\langle \Delta p^\alpha \rangle}{\Delta t} = -Y^\alpha \tag{2.2b}$$

As such, $\hat{q}$ and $\mathrm{d}E/\,\mathrm{d}x$ may be obtained from $X^{\alpha\beta}$ and Y^α:

$$\hat{q} = \frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right) X^{\alpha\beta} \tag{2.3a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = \frac{v^\alpha}{v}\frac{\langle \Delta p^\alpha \rangle}{\Delta t} = -\frac{v^\alpha}{v} Y^\alpha \tag{2.3b}$$

Should these auxiliary calcs be moved to an appendix? To prove the first equality of eq. 2.3a,

$$\frac{1}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)\frac{\langle \Delta p^\alpha \Delta p^\beta \rangle}{\Delta t} = \frac{1}{\Delta x}\left\langle (\Delta\mathbf{p})^2 - \left(\frac{\mathbf{v}\cdot\Delta\mathbf{p}}{v^2}\right)\right\rangle = \frac{\langle (\Delta\mathbf{p}_T)^2 \rangle}{\Delta x}$$

To prove the second equality of eq. 2.3a, eq. 2.2a is plugged in and the symmetry of $(\delta^{\alpha\beta} - v^\alpha v^\beta/v^2)$ under the switch of α and β is used.

[1] Requires further explanation if the event-averages haven't been explained so far.

To prove the first equality of eq. 2.3b, it is easiest to start from the definition of the stopping power:

$$\frac{\langle \Delta E\rangle}{\Delta x} = \frac{1}{v\Delta t}\left\langle \Delta(\sqrt{\mathbf{p}^2+m^2})\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\Delta(\mathbf{p}^2)}{2\sqrt{\mathbf{p}^2+m^2}}\right\rangle = \frac{1}{v\Delta t}\left\langle \frac{\mathbf{p}\cdot\Delta\mathbf{p}}{\sqrt{p^2+m^2}}\right\rangle = \frac{v^\alpha}{v}\frac{\Delta p^\alpha}{\Delta t}$$

$X^{\alpha\beta}$ is obtained using the event-averaged field correlators previously derived, while determining Y^α requires calculating the event-average of the correlator of one of the fields and each event's deviation of the quark-probe density from the event-averaged density, n, a quantity which is difficult to access.

In order to circumvent the challenges of calculating Y^α, it is assumed that the Fokker-Planck equation should, if the system were in equilibrium at a temperature T, admit the Boltzmann distribution function as a solution:

$$n^{\text{eq}}(\mathbf{p}) \propto \exp(E_\mathbf{p}/T), \qquad \text{with } E_\mathbf{p} = \sqrt{\mathbf{p}^2 + {m_Q}^2}$$

with m_Q the mass of the quark probe. This assumption allows obtaining Y^α from $X^{\alpha\beta}$ via the simple relation:

$$Y^\alpha = \frac{v^\beta}{T}X^{\alpha\beta} \tag{2.4}$$

For the purpose of obtaining Y^α from $X^{\alpha\beta}$, T is taken to be the temperature of a thermalized QGP with an energy density equal to that of the glasma[2]. That is, the temperature which would characterize the system achieved if the glasma were allowed to reach equilibrium without expanding. [Comments on validity?] - partly discussed in sec. 5 of Mrowczyncki_2018

From eqs. 2.3 and 2.4, the final expressions for determining $\hat{q}$ and $\mathrm{d}E/\mathrm{d}x$ from $X^{\alpha\beta}$ become:[3]

$$\hat{q} = \frac{2}{v}\left(\delta^{\alpha\beta} - \frac{v^\alpha v^\beta}{v^2}\right)X^{\alpha\beta} \tag{2.5a}$$

$$\frac{\mathrm{d}E}{\mathrm{d}x} = -\frac{v}{T}\frac{v^\alpha v^\beta}{v^2}X^{\alpha\beta} \tag{2.5b}$$

As such, obtaining $\hat{q}$ and $\mathrm{d}E/\mathrm{d}x$ requires calculating the $\delta^{\alpha\beta}X^{\alpha\beta}$ and $v^\alpha v^\beta X^{\alpha\beta}$ projections of the $X^{\alpha\beta}$ tensor.

2.2 The $X^{\alpha\beta}$ tensor

The first collision term is obtained from the correlators of chromodynamic fields:[4]

$$\begin{aligned} X^{\alpha\beta}(t,\mathbf{x},\mathbf{v}) &= \frac{1}{N_c}\int_0^t \mathrm{d}t'\ \text{Tr}_c\left[\left\langle F^\alpha(t,\mathbf{x})F^\beta(t',\mathbf{y})\right\rangle\right] \\ &= \frac{1}{2N_c}\int_0^t \mathrm{d}t'\ \left\langle F_a^\alpha(t,\mathbf{x})F_a^\beta(t',\mathbf{y})\right\rangle \end{aligned} \tag{2.6}$$

[2] The derivation of the glasma's energy density should end up before this section, so the reader will be familiar with the concept.
[3] The final form of dEdx will depend on wheteher the projection calculated will be on va*vb or va*vb/v**2
[4] The first line of this equation disagrees with [7] by a fator of $1/2$ but agrees with [8]. However, the second line is in agreement with [7]. This is presumably due to a typo in [7].

where N_c is the number of colors, Tr_c is a color trace in the fundamental representation, $\mathbf{y} = \mathbf{x} - \mathbf{v}(t - t')$, $F^\alpha \equiv F_a^\alpha t^a$[5] is the α component of the Lorentz color force, and a are color indices in the adjoint representation, which run from 1 through $N_c{}^2 - 1$. Thus, the $X^{\alpha\beta}$ component of the tensor is obtained by integrating, in time, over the full past trajectory of the probe, the correlator of the α and β components of the Lorentz color force; the first at the "present" (time t, position $\mathbf{x}$), and the second at a past point of the probe's trajectory (time t', position $\mathbf{y}$).

The Lorentz color force a quark is subject to when traveling with velocity $\mathbf{v}$ at time t and position $\mathbf{x}$ is:

$$\mathbf{F}_a(t, \mathbf{x}) = g\big(\mathbf{E}_a(t, \mathbf{x}) + \mathbf{v} \times \mathbf{B}_a(t, \mathbf{x})\big) \tag{2.7a}$$

$$F_a^\alpha(t, \mathbf{x}) = g\big(E_a^\alpha(t, \mathbf{x}) + \varepsilon^{\alpha\gamma\gamma'} v^\gamma B_a^{\gamma'}(t, \mathbf{x})\big) \tag{2.7b}$$

where g is the strong coupling constant and $\varepsilon^{\alpha\beta\gamma}$ is the Levi-Civita or antisymmetric tensor with $\varepsilon^{123} = 1$.

Plugging eq. 2.7b into eq. 2.6 allows writing the unfolded expression for $X^{\alpha\beta}$:

$$\begin{aligned} X^{\alpha\beta}(t, \mathbf{x}, \mathbf{v}) = \frac{g^2}{2N_c} \int_0^t \mathrm{d}t' \Big[& \langle E_a^\alpha(t, \mathbf{x}) E_a^\beta(t - t', \mathbf{y}) \rangle \\ & + \varepsilon^{\beta\gamma\gamma'} v^\gamma \left\langle E_a^\alpha(t, \mathbf{x}) B_a^{\gamma'}(t - t', \mathbf{y}) \right\rangle \\ & + \varepsilon^{\alpha\gamma\gamma'} v^\gamma \left\langle B_a^{\gamma'}(t, \mathbf{x}) E_a^\beta(t - t', \mathbf{y}) \right\rangle \\ & + \varepsilon^{\alpha\gamma\gamma'} \varepsilon^{\beta\delta\delta'} v^\gamma v^\delta \left\langle B_a^{\gamma'}(t, \mathbf{x}) B_a^{\delta'}(t - t', \mathbf{y}) \right\rangle \Big] \end{aligned} \tag{2.8}$$

[5] A proper appendix or introduction to SU(Nc) will be necessary

Bibliography

[1] S. Navas et al. Review of particle physics. *Phys. Rev. D*, 110(3):030001, 2024. doi: 10.1103/PhysRevD.110.030001.

[2] S. Bailey, T. Cridge, L. A. Harland-Lang, A. D. Martin, and R. S. Thorne. Parton distributions from lhc, hera, tevatron and fixed target data: Msht20 pdfs. *The European Physical Journal C*, 81(4), Apr. 2021. ISSN 1434-6052. doi: 10.1140/epjc/s10052-021-09057-0. URL http://dx.doi.org/10.1140/epjc/s10052-021-09057-0.

[3] J. Albacete and C. Marquet. Gluon saturation and initial conditions for relativistic heavy ion collisions. *Progress in Particle and Nuclear Physics*, 76:1–42, May 2014. ISSN 0146-6410. doi: 10.1016/j.ppnp.2014.01.004. URL http://dx.doi.org/10.1016/j.ppnp.2014.01.004.

[4] W. van der Schee. Gravitational collisions and the quark-gluon plasma, 2014. URL https://arxiv.org/abs/1407.1849.

[5] M. E. Carrington, A. Czajka, and S. Mrówczyński. The energy-momentum tensor at the earliest stage of relativistic heavy-ion collisions. *The European Physical Journal A*, 58(1):5, 2022. ISSN 1434-601X. doi: 10.1140/epja/s10050-021-00600-x. URL https://doi.org/10.1140/epja/s10050-021-00600-x.

[6] F. Gelis. The initial stages of high energy heavy ion collisions. *Institut de Physique Théorique (IPhT) Habilitation Thesis*, Dec. 2011.

[7] M. E. Carrington, A. Czajka, and S. Mrówczyński. Transport of hard probes through glasma. *Physical Review C*, 105(6), June 2022. ISSN 2469-9993. doi: 10.1103/physrevc.105.064910. URL http://dx.doi.org/10.1103/PhysRevC.105.064910.

[8] S. Mrówczyński. Heavy quarks in turbulent QCD plasmas. *The European Physical Journal A*, 54(3):43, 2018. ISSN 1434-601X. doi: 10.1140/epja/i2018-12478-5. URL https://doi.org/10.1140/epja/i2018-12478-5.

Light-Cone Coordinates

May need to be reviewed so it has a similar structure and flow to the Bjorken coordinates appendix.

Throughout the derivations, it will be useful to work in light-cone coordinates. A four-vector $a^\mu = (a^0, a^1, a^2, a^3)$ may be expressed in light-cone coordinates as

$$a^\mu = (a^+, a^-, a^2, a^3) \equiv (a^+, a^-, \mathbf{a}_\perp), \qquad \mu = (+, -,\ 2,\ 3) \tag{9}$$

with

$$a^+ = \frac{a^0 + a^1}{\sqrt{2}} \qquad a^- = \frac{a^0 - a^1}{\sqrt{2}} \qquad \mathbf{a}_\perp = (a^2, a^3) \tag{10}$$

The dot product between two four-vectors a^μ and b^ν becomes

$$a \cdot b = a^+ b^- + a^- b^+ - \mathbf{a}_\perp \cdot \mathbf{b}_\perp = a^+ b^- + a^- b^+ - a^2 b^2 - a^3 b^3 \tag{11}$$

the metric being

$$g_{\mu\nu} = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -1 \end{bmatrix} \tag{12}$$

The components of the corresponding covariant four-vector a_μ are

$$a_\mu = (a_+, a_-, a_2, a_3) = (a^-, a^+, -a^2, -a^3) \tag{13}$$

Considering the position four-vector,

$$x^\mu = (x^+, x^-, x^2, x^3) = \left(\frac{x^0 + x^1}{\sqrt{2}}, \frac{x^0 - x^1}{\sqrt{2}}, x^2, x^3 \right) \tag{14}$$

we may obtain the new base vectors. Focusing on the first two coordinates:

$$\vec{r} = x^0 \vec{e}_0 + x^1 \vec{e}_1, \quad \text{with} \quad \vec{e}_0 = (1, 0, 0, 0), \quad \vec{e}_1 = (0, 1, 0, 0) \tag{15}$$

$$\begin{aligned} \vec{e}_+ &= \frac{\partial \vec{r}}{\partial x^+} = \frac{1}{\sqrt{2}} \vec{e}_0 + \frac{1}{\sqrt{2}} \vec{e}_1 \\ \vec{e}_- &= \frac{\partial \vec{r}}{\partial x^-} = \frac{1}{\sqrt{2}} \vec{e}_0 - \frac{1}{\sqrt{2}} \vec{e}_1 \end{aligned} \tag{16}$$

Allowing us to rewrite $\vec{r}$:

$$\vec{r} = x^+ \vec{e}_+ + x^- \vec{e}_- \tag{17}$$

Thus, the x^+ and x^- axes lie on the light cone (Fig. 1).

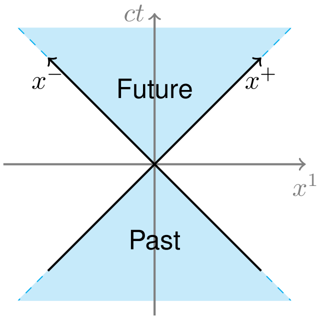

Figure 1: Position four-vector axes in light-cone coordinates.

As for the four-gradient,

$$\partial_\mu = (\partial_+, \partial_-, \partial_2, \partial_3) = \left(\frac{\partial}{\partial x^+}, \frac{\partial}{\partial x^-}, \frac{\partial}{\partial x^2}, \frac{\partial}{\partial x^3}\right) \tag{18}$$

$$\partial^\mu = (\partial^+, \partial^-, \partial^2, \partial^3) = (\partial_-, \partial_+, -\partial_2, -\partial_3) \tag{19}$$

Abusing language and the notation, one may similarly define the Dirac matrices “in light-cone coordinates”.

$$\gamma^+ = \frac{\gamma^0 + \gamma^1}{\sqrt{2}} \qquad \gamma^- = \frac{\gamma^0 - \gamma^1}{\sqrt{2}} \tag{20}$$

They obey anticommutation relations of the same form as the usual Dirac matrices, with the new metric of Eq. 12.

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu}, \qquad \mu, \nu = (+, -, 2, 3) \tag{21}$$

Bjorken/Milne[6] Coordinates

In this system, two familiar quantities are used as the first two coordinates: proper time, τ of an object with $\mathbf{x}_\perp = 0$? which set off from the origin with constant velocity? – which is invariant under boosts along the longitudinal direction, (x^1) –, and rapidity of some body in some particular circumstances too, η. The transverse coordinates are unchanged, as in light-cone coordinates.

$$\begin{aligned} \tau &= \sqrt{(x^0)^2 - (x^1)^2} = \sqrt{2x^+x^-} \\ \eta &= \frac{1}{2}\log\left(\frac{x^0+x^1}{x^0-x^1}\right) = \frac{1}{2}\log\left(\frac{x^+}{x^-}\right) \end{aligned} \tag{22}$$

As for the inverse transformation,

$$x^+ = \frac{\tau}{\sqrt{2}}e^{\eta} \qquad x^- = \frac{\tau}{\sqrt{2}}e^{-\eta} \tag{23}$$

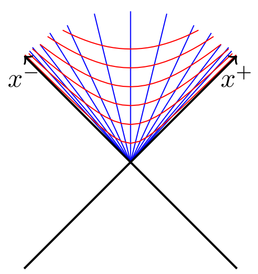

Figure 2: Contour lines for proper time, τ, in red, and rapidity, η, in blue, in the first quadrant.

We may find the new basis vectors: versores n deviam ser "e"

$$\vec{r} = x^+\vec{e}_+ + x^-\vec{e}_-$$

$$\begin{aligned} \vec{e}_\tau &= \frac{\partial \vec{r}}{\partial \tau} = \frac{e^{\eta}}{\sqrt{2}}\vec{e}_+ + \frac{e^{-\eta}}{\sqrt{2}}\vec{e}_- = \frac{x^+}{\tau}\vec{e}_+ + \frac{x^-}{\tau}\vec{e}_- \\ \vec{e}_\eta &= \frac{\partial \vec{r}}{\partial \eta} = \frac{\tau}{\sqrt{2}}e^{\eta}\vec{e}_+ - \frac{\tau}{\sqrt{2}}e^{-\eta}\vec{e}_- = x^+\vec{e}_+ - x^-\vec{e}_- \end{aligned} \tag{24}$$

which are not, in general, of unit length.

By requiring

$$\vec{A} = A^+\vec{e}_+ + A^-\vec{e}_- \equiv A^\tau\vec{e}_\tau + A^\eta\vec{e}_\eta,$$

then

$$\begin{aligned} A^\tau &= \frac{x^-A^+ + x^+A^-}{\tau} \\ A^\eta &= \frac{x^-A^+ - x^+A^-}{\tau^2} \end{aligned} \tag{25}$$

[6] what Margaret calls them on page 12

Particularly,

$$\vec{r} = \tau \vec{e}_\tau \tag{26}$$